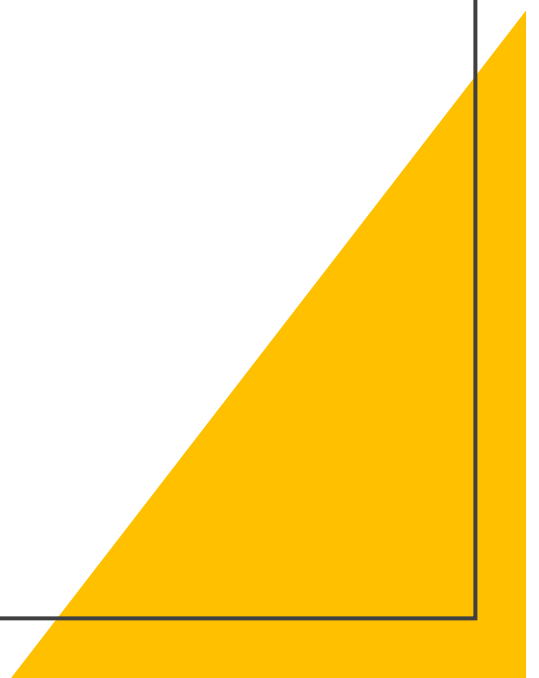


# Relativistic Effects & Solvation Models

Lecture 6

Liviu Ungur



# Lecture ( $\approx 1$ h)

- Wave function analysis
- Atomic charges
- Relativistic corrections
- Solvation Models

# Electron density distribution

The total electron density can be obtained as a sum of probability distributions of each individual electron:

$$\rho(\vec{r}) = \sum_{i=1}^N |\chi_i(\vec{r})|^2 \quad \chi_i(\vec{r}) = \sum_{\mu=1}^K c_{\mu i} \phi_{\mu}(\vec{r})$$

Where  $N$  is the number of electrons in the system and  $K$  is the number of functions in the basis. Therefore the total density can be expressed as:

$$\rho(\vec{r}) = \sum_{\mu, \nu=1}^K \sum_{i=1}^N c_{\mu i}^* c_{\nu i} \phi_{\mu}^*(\vec{r}) \phi_{\nu}(\vec{r})$$

Integrating this over all space gives the total number of electrons in the system:

$$N = \int \rho(\vec{r}) d^3\vec{r} = \sum_{\mu, \nu=1}^K \sum_{i=1}^N c_{\mu i}^* c_{\nu i} \int \phi_{\mu}^*(\vec{r}) \phi_{\nu}(\vec{r}) d^3\vec{r}$$

Two shorthands are commonly defined in this expression:

$$P_{\mu\nu} = \sum_{i=1}^N c_{\mu i}^* c_{\nu i} \quad S_{\mu\nu} = \int \phi_{\mu}^*(\vec{r}) \phi_{\nu}(\vec{r}) d^3\vec{r} \quad N = \sum_{\mu, \nu=1}^K P_{\mu\nu} S_{\mu\nu}$$

"density matrix"                      "overlap matrix"

# Electron density distribution

We can now find out how much a particular pair of orbitals contributes to the total observable electron density:

$$N_{\mu\nu} = \sum_{i=1}^N \underbrace{c_{\mu i}^* c_{\nu i}}_{\text{orbital populations}} \underbrace{\int \varphi_{\mu}^*(\vec{r}) \varphi_{\nu}(\vec{r}) d^3\vec{r}}_{\text{orbital overlap}} = P_{\mu\nu} S_{\mu\nu}$$

This density partitioning into pairs of orbitals is useful in charge and bonding structure analysis. In atom-centred basis sets there are two possible situations:

1. Both  $\varphi_{\mu}$  and  $\varphi_{\nu}$  are located on the same atom. All of the density in  $N_{\mu\nu}$  can be assigned to this atom and interpreted as contributing to its charge.
2. The two orbitals  $\varphi_{\mu}$  and  $\varphi_{\nu}$  are located on different atoms. The density in  $N_{\mu\nu}$  is then shared between the two atoms and can be interpreted as contributing to a chemical bond as well as charges.

This formalism can be generalized to partition spin as well as charge:

$$P_{\mu\nu}^{\alpha} = \sum_{i=1}^N c_{\mu i}^{\alpha*} c_{\nu i}^{\alpha} \quad P_{\mu\nu}^{\beta} = \sum_{i=1}^N c_{\mu i}^{\beta*} c_{\nu i}^{\beta}$$

# Mulliken populations and charges

Approximate atomic charges can be derived from the populations of specific atomic orbitals using the procedure suggested by Mulliken. For each atom in the system:

$$Q_A = Z_A - \sum_{\mu \in A} \sum_{\nu \in A} P_{\mu\nu} S_{\mu\nu} - \frac{1}{2} \sum_{\mu \in A} \sum_{\nu \in B} P_{\mu\nu} S_{\mu\nu}$$

This can be expanded as the following set of rules:

1. Start with the charge of the bare nucleus.
2. If the basis functions  $\phi_\mu$  and  $\phi_\nu$  are both on atom A, assign all the corresponding density to the charge of atom A.
3. If the basis function  $\phi_\mu$  is on atom A and  $\phi_\nu$  is on atom B, assign half of the corresponding density to the charge of atom A.

Summation without sign and nuclear charge gives *atomic populations*, and summation with spin included gives *atomic spin densities*.

Useful features of Mulliken populations:

1. Qualitatively correct (no quantitative method exists anyway).
2. Very cheap to calculate (ground state property).
3. Show correct trends in chemical reactions.

# Löwdin populations and charges

Mulliken populations have a few problems:

1. Diagonal elements of  $P_{\mu\nu}$  may be larger than two, implying more than two electrons in an orbital (Pauli principle violation).
2. Off-diagonal elements of  $P_{\mu\nu}$  may be negative, implying a negative number of electrons in an orbital.
3. Equal partitioning of electron density between participating atoms is without any foundation.
4. Diffuse functions centred on atom  $A$  are still counted as only belonging to  $A$ .
5. Mulliken charges do not reproduce the multipole moments of the molecule.

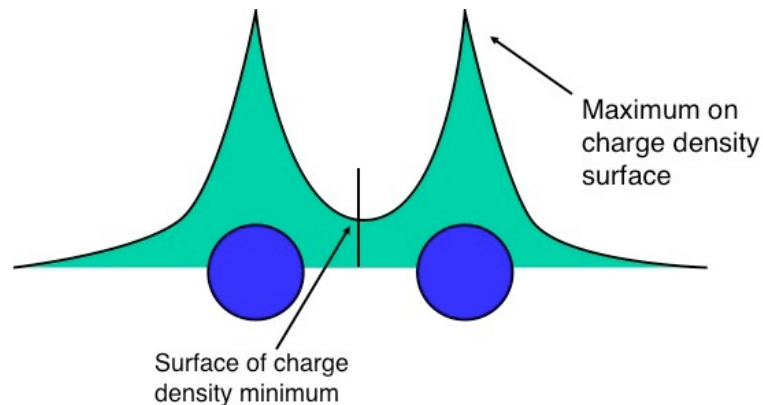
A refinement (at least to points 1-3) was suggested by Löwdin, who used the cyclic permutation property of the trace

$$N = \sum_{\mu,\nu=1}^K P_{\mu\nu} S_{\mu\nu} = \text{Tr}(\mathbf{PS}) = \text{Tr}(\mathbf{S}^{1/2} \mathbf{PS}^{1/2}) \quad N_{\mu\nu} = [\mathbf{S}^{1/2} \mathbf{PS}^{1/2}]_{\mu\nu}$$

to point out that orthogonalized basis orbitals may be used for the population analysis instead of the (possibly degenerate) original basis set.

# Bader Atomic Charges

**R. Bader** developed an intuitive way of dividing molecules into atoms called the Quantum Theory of Atoms in Molecules (QTAIM). His definition of an atom is based purely on the electronic charge density. **Bader uses** what are called **zero flux surfaces to divide atoms**. A zero-flux surface is a 2-D surface on which the charge density is a minimum perpendicular to the surface. Typically in molecular systems, the charge density reaches a minimum between atoms and this is a natural place to separate atoms from each other.



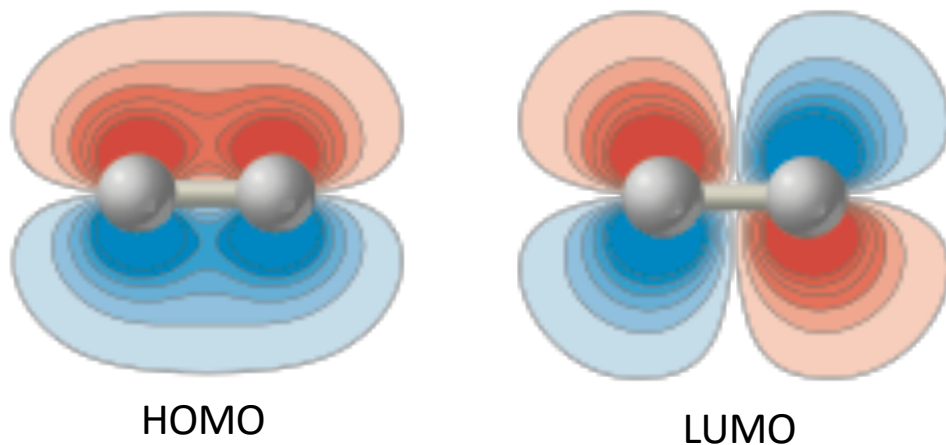
Bader's theory of **atoms in molecules** is often useful for charge analysis. For example, the charge enclosed within the Bader volume is a good approximation to the total electronic charge of an atom. The charge distribution can be used to determine multipole moments of interacting atoms or molecules. Bader's analysis has also been used to define the hardness of atoms, which can be used to quantify the cost of removing charge from an atom. The theory also provides a definition for chemical bonding that gives numerical values for bond strength.

# Hirshfeld Atomic Charges

A general and natural choice is to share the charge density at each point between the atoms, in proportion to their free-atom densities at the corresponding distances from the nuclei.

Hirshfeld charges ([Hirshfeld, 1977](#)) are defined relative to the **deformation density**, this is the difference between the *molecular (relaxed)* and *unrelaxed* atomic charge densities.

**Deformation density** appears as a natural result of chemical bonding and sharing of electrons between atoms.



# Mulliken bond order

The bond order between atom A and atom B is defined as sum over all such electron density as has the basis function  $\varphi_\mu$  is on atom A and  $\varphi_\nu$  is on atom B:

$$q_{AB} = 2 \sum_{\mu \in A} \sum_{\nu \in B} P_{\mu\nu} S_{\mu\nu}$$

(a similar alteration to the one above may be made to get the Löwdin bond order).

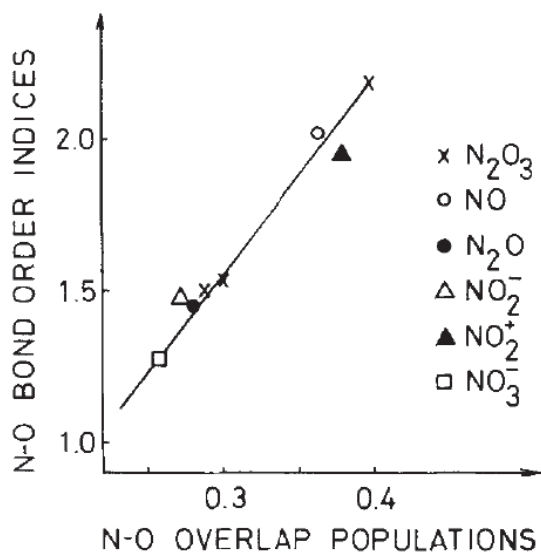


TABLE IV. Bond orders for propylene, acetaldehyde, and acetone.

| Propylene                          |      | Acetaldehyde                       |      | Acetone                            |      |
|------------------------------------|------|------------------------------------|------|------------------------------------|------|
| C <sub>1</sub> -H <sub>1</sub>     | 0.92 | C <sub>1</sub> -H <sub>1</sub>     | 0.91 | C <sub>1</sub> -H <sub>1</sub>     | 0.89 |
| C <sub>1</sub> -H <sub>2</sub>     | 0.90 | C <sub>1</sub> -H <sub>2</sub>     | 0.90 | C <sub>1</sub> -H <sub>2</sub>     | 0.88 |
| C <sub>1</sub> -H <sub>3</sub>     | 0.90 | C <sub>1</sub> -H <sub>3</sub>     | 0.90 | C <sub>1</sub> -H <sub>3</sub>     | 0.88 |
| C <sub>1</sub> -C <sub>2</sub> (σ) | 0.95 | C <sub>1</sub> -C <sub>2</sub> (σ) | 0.94 | C <sub>1</sub> -C <sub>2</sub> (σ) | 0.89 |
| (π)                                | 0.11 | (π)                                | 0.11 | (π)                                | 0.10 |
| (π)                                | 0.06 | (π)                                | 0.05 | (π)                                | 0.04 |
|                                    | 1.13 |                                    | 1.10 |                                    | 1.04 |
| C <sub>2</sub> -C <sub>3</sub> (σ) | 0.95 | C <sub>2</sub> -O(σ)               | 0.52 | C <sub>2</sub> -O(σ)               | 0.52 |
| (π)                                | 0.98 | (π)                                | 0.96 | (π)                                | 0.94 |
| (π)                                | 0.11 | (π)                                | 0.13 | (π)                                | 0.04 |
|                                    | 2.04 |                                    | 1.61 |                                    | 1.49 |
| C <sub>2</sub> -H <sub>4</sub>     | 0.88 | C <sub>2</sub> -H <sub>4</sub>     | 0.86 |                                    |      |
| C <sub>3</sub> -H <sub>5</sub>     | 0.93 |                                    |      |                                    |      |
| C <sub>3</sub> -H <sub>6</sub>     | 0.91 |                                    |      |                                    |      |

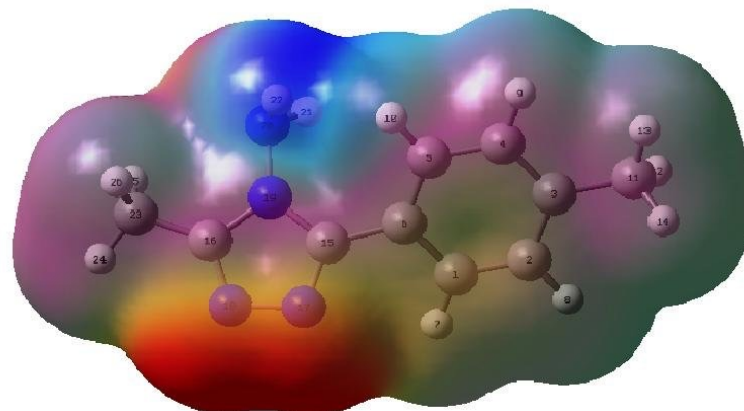
# Molecular Electrostatic Potential

The molecular electrostatic potential (MEP) is useful in describing the **charge distribution** of the valence electrons in a molecule.

$$\phi(x_1, y_1, z_1) = \sum_i \frac{q_i}{4\pi\epsilon_0 r_i} - \iiint \frac{\rho(x, y, z)}{4\pi\epsilon_0 r_e} dx dy dz$$

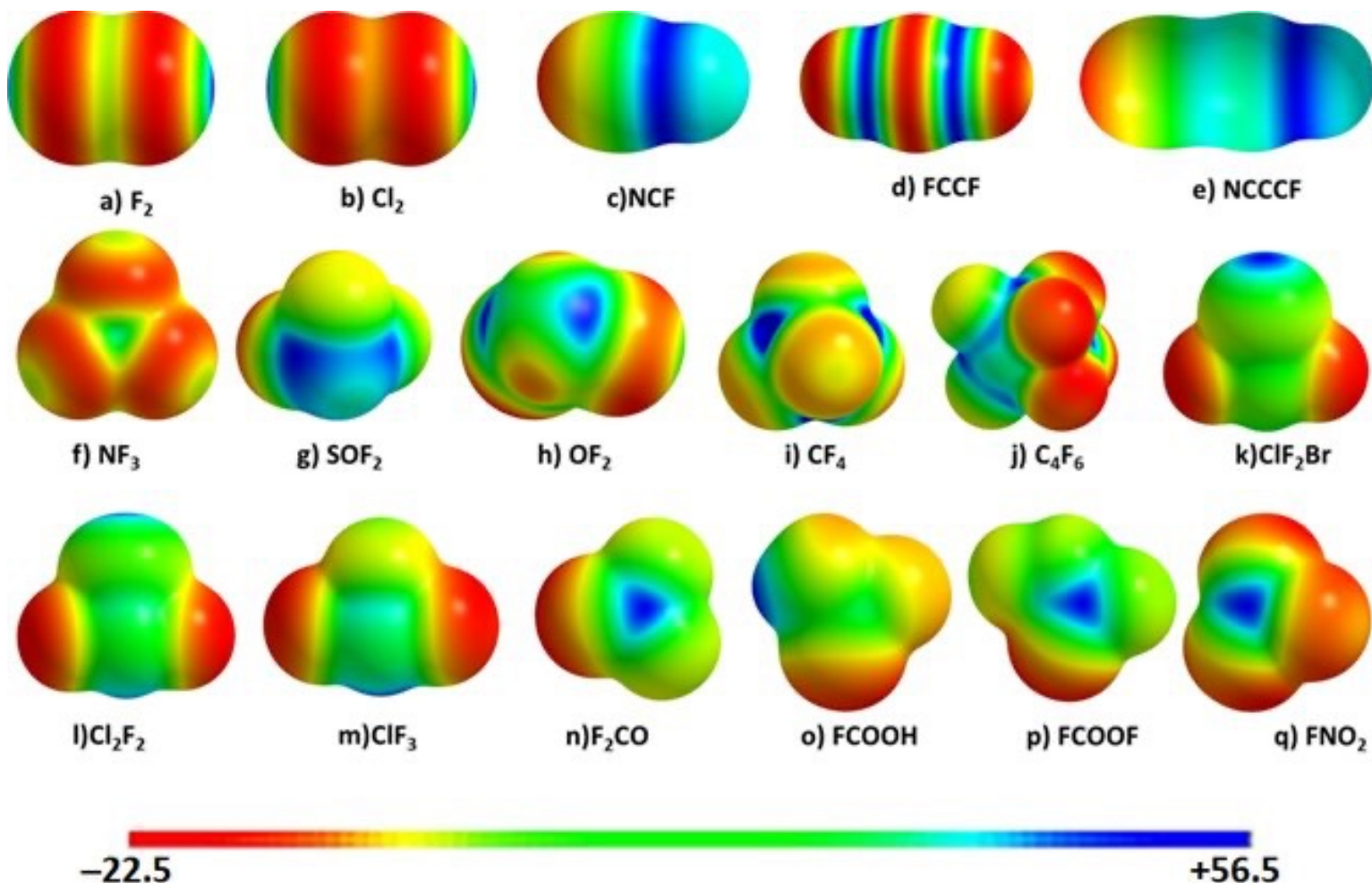
nuclear term  
(positive)

electronic term  
(negative)



**Surface grid of constant electron density for the HF molecule.** The fluorine atom is shown in green. The colour shading on the grid indicates the value of the molecular electrostatic potential. Red and blue correspond to negative and positive values, respectively.

# Molecular Electrostatic Potential



# Molecular orbitals and orbital localization

MOs as such are mostly used for building the various integrals involved in post-HF energy and property calculations. They can be plotted, and the plots look pretty.

## Localization

The Slater determinant is invariant under an arbitrary unitary transformation of molecular orbitals – they may be mixed the way we see fit.

Boys' criterion: maximize the distances between orbital centroids.

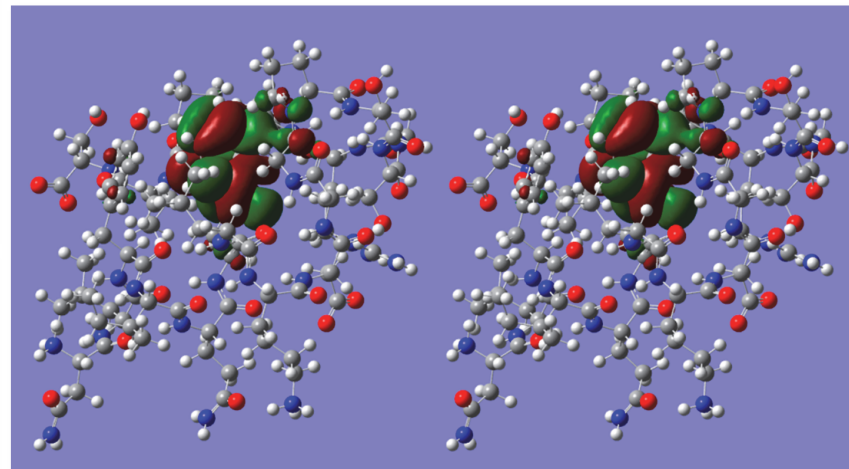
$$\{\phi_i^{\text{loc}}\} = \arg \max \left( \sum_{ij} \left[ \langle \phi_i | \vec{r} | \phi_i \rangle - \langle \phi_j | \vec{r} | \phi_j \rangle \right]^2 \right)$$

Boys method is equivalent to minimizing the orbital dispersion.

Edminston-Ruedenberg criterion: maximize the self-repulsion energy.

$$\{\phi_i^{\text{loc}}\} = \arg \max \left( \sum_i \langle \phi_i \phi_i | r^{-1} | \phi_i \phi_i \rangle \right)$$

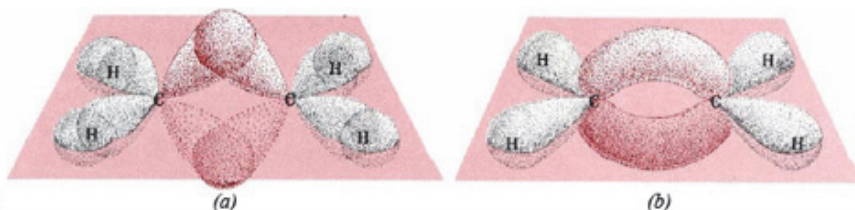
The results of Boys' and ER methods are generally similar.



Stereo plot of HOMO isosurface in a Trp-cage mini-protein.

# Molecular orbitals and orbital localization

*N.B.* Although they still produce the same Slater determinant, the localized orbitals are no longer eigenfunctions of the SCF operator – there is no such thing as the “energy of a localized orbital”.



Banana bonds – a common orbital localization artefact.

Pipek-Mezey criterion: maximize the sum of density squares on all atoms.

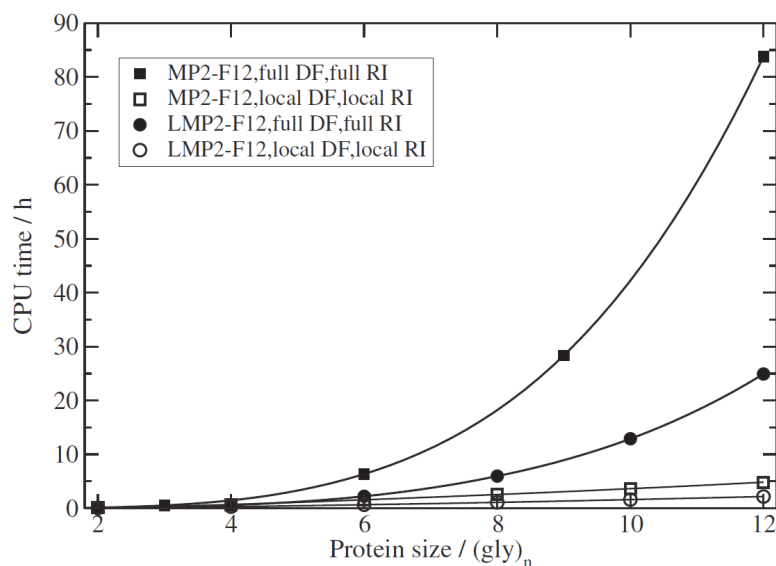
$$\{\phi_i^{\text{loc}}\} = \arg \max \left( \sum_n |\rho_n|^2 \right)$$

PM localization preserves the separation between  $\sigma$ - and  $\pi$ -orbitals in  $\pi$ -bonded systems, such as ethylene (see the table on the right).

| Canonical MOs | Pipek-Mezey LMOs | Boys LMOs |
|---------------|------------------|-----------|
|               |                  |           |
|               |                  |           |
|               |                  |           |
|               |                  |           |
|               |                  |           |
|               |                  |           |

# HF and post-HF methods with localized orbitals

Orbital localization reduces the extent of overlap between canonical orbitals and therefore increases the number of zero and near-zero two-electron integrals. Formulating QC methods in terms of *localized* orbitals improves their efficiency.



Approximation 1: excitations are restricted to be local.

Approximation 2: correlation between distant pairs of local orbitals is ignored.

Both make physical sense – it is known that electron correlation decays rapidly with distance.

Typically >95% of the correlation energy (see below) is preserved.

| Correlation energy <sup>a</sup> |                  |            |                    |            |                    |                          |                              |
|---------------------------------|------------------|------------|--------------------|------------|--------------------|--------------------------|------------------------------|
| Molecule                        | $E_{\text{SCF}}$ | MP2(full)  | MP2(local)         | CCSD(full) | CCSD(local)        | CCSD(local) <sup>b</sup> | CCSD/MP2(local) <sup>c</sup> |
| VDZ basis: <sup>d</sup>         |                  |            |                    |            |                    |                          |                              |
| Propane                         | −118.270 461     | −0.446 239 | −0.436 124 (97.73) | −0.496 129 | −0.485 978 (97.95) | −0.486 150 (97.99)       | −0.489 501 (98.66)           |
| Butane                          | −157.306 738     | −0.589 737 | −0.575 103 (97.52) | −0.652 361 | −0.637 837 (97.77) | −0.638 109 (97.82)       | −0.643 409 (98.63)           |
| Pentane                         | −196.342 887     | −0.733 296 | −0.714 074 (97.38) | −0.808 656 | −0.789 709 (97.66) | −0.790 081 (97.70)       | −0.797 321 (98.60)           |

# Relativistic effects

Einstein's relativity theory postulated that:

- **Speed of light  $c$  is constant** in any system of in all inertia systems ( $\sim 137.036 a. u.$ )
- **All physical laws are the same** in all inertia system →
- **Space coordinates  $(x,y,z)$  and time  $(t)$**  are four coordinates of a single space-time framework, and therefore, **are equivalent to each other.**
- **Lorentz transformations** allow to relate the coordinates between two inertia systems

$$\left[ -\frac{1}{2m} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V \right] \Psi = i \frac{\partial \Psi}{\partial t} \quad \rightarrow \quad \text{Schrodinger equation is NOT relativistically correct: space coordinates are described by second derivative, while time – at first derivative.}$$

$$m = m_0 \left( \sqrt{1 - \frac{v^2}{c^2}} \right)^{-1} \quad \rightarrow \quad \text{the **mass** of the fast-moving electrons (e.g. 1s) is higher than the mass of valence electrons}$$

This leads to a contraction of the 1s orbital, which in turn, leads to a “lower effective nuclear charge” felt by outer electrons → outer orbitals get slightly more diffuse.

# Dirac Equation

Dirac developed the **relativistic quantum chemistry**. He proposed (derived) the following equation instead of the Schrodinger equation:

$$[c\boldsymbol{\alpha} \cdot \mathbf{p} + \beta mc^2]\Psi = i\frac{\partial \Psi}{\partial t}$$

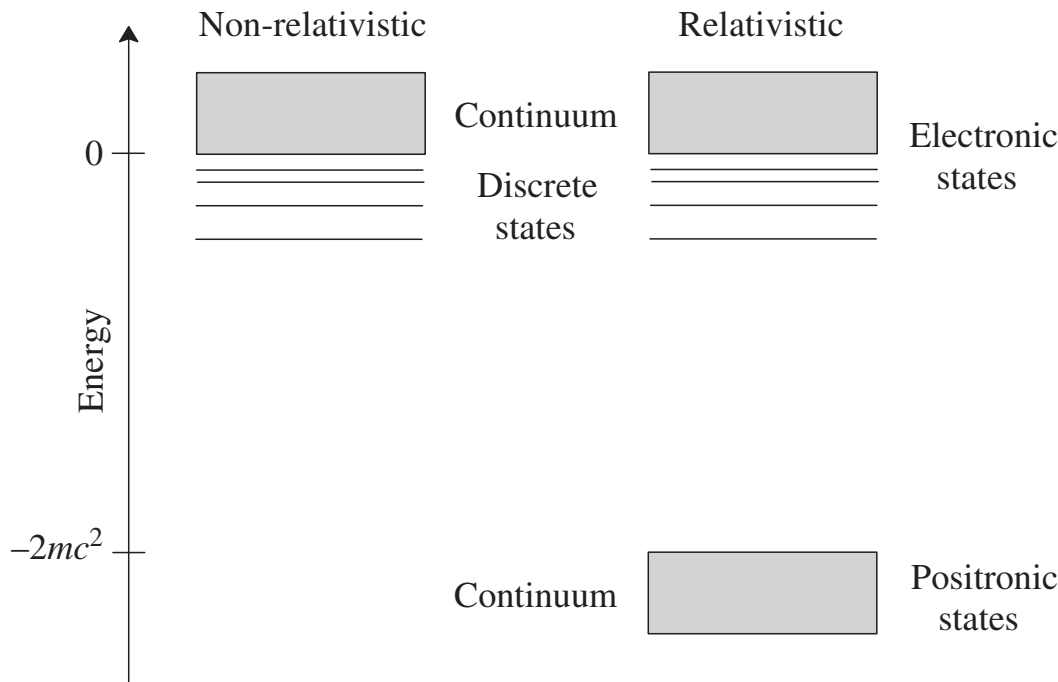
where:

$$\boldsymbol{\alpha}_{x,y,z} = \begin{pmatrix} 0 & \boldsymbol{\sigma}_{x,y,z} \\ \boldsymbol{\sigma}_{x,y,z} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} \mathbf{I} & 0 \\ 0 & -\mathbf{I} \end{pmatrix}$$
$$\boldsymbol{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Solution of the Dirac equation are 4-component wave functions:

$$\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = \begin{pmatrix} \Psi_L \\ \Psi_S \end{pmatrix}$$

# Energy spectra from Dirac Equation



$$mc^2 \approx 0.511 \text{ MeV}$$

Binding energy of  
electron in H:  
 $1.36 \times 10^{-5} \text{ MeV}$

To align the relativistic  
and non-relativistic  
energies, the following  
term may be used  
instead of  $\beta$  :

$$\beta' = \begin{pmatrix} 0 & 0 \\ 0 & -2\mathbf{I} \end{pmatrix}$$

Non-relativistic and relativistic solutions. The energy scale of the discrete states is exaggerated relative to the energy separation of the continuum states.

# Energy spectra from Dirac Equation

The four-component Dirac equation can be transformed such as to compute only the  $\Psi_L$  (large component, the solution for electrons). The transformed Hamiltonian becomes (Pauli-Dirac equation):

$$\left[ \underbrace{\frac{\mathbf{p}^2}{2m} + \mathbf{V}}_{\text{Kinetic and potential energy}} - \underbrace{\frac{\mathbf{p}^4}{8m^3c^2}}_{\text{mass-velocity term}} + \underbrace{\frac{Z\mathbf{s} \cdot \mathbf{l}}{2m^2c^2r^3}}_{\text{spin-orbit interaction}} + \underbrace{\frac{Z\pi\delta(\mathbf{r})}{2m^2c^2}}_{\text{Darwin correction}} \right] \Psi_L = E\Psi_L$$

**Scalar relativistic corrections** = “mass-velocity” + “Darwin” term

$\Psi_L$  = is a 2-component wave function (spinor)

**Douglas-Kroll Hess** is a transformation of the 2-component Dirac Hamiltonian into the non-relativistic + various correction terms

# Relativistic electrostatic interactions

Dirac operator for one electron (kinetic energy term):

$$c\boldsymbol{\alpha} \cdot (\mathbf{p} + \mathbf{A}) + \beta' mc^2$$

$\mathbf{A}$  is the vector potential (e.g. electric or magnetic fields),  $\mathbf{p}$  is momentum.

Many-electron operator of kinetic energy of all electrons is a sum of such operators.

Dirac operator incorporates relativistic effects for the kinetic energy. In order to describe atomic and molecular systems, the potential energy operator must also be modified.

$$\mathbf{V}(\mathbf{r}_{12}) = \frac{q_1 q_2}{r_{12}}$$

non-relativistic  
potential energy

$$\mathbf{V}_{ee}^{\text{Coulomb-Breit}}(\mathbf{r}_{12}) = \frac{1}{r_{12}} - \frac{1}{2r_{12}} \left[ \boldsymbol{\alpha}_1 \cdot \boldsymbol{\alpha}_2 + \frac{(\boldsymbol{\alpha}_1 \cdot \mathbf{r}_{12})(\boldsymbol{\alpha}_2 \cdot \mathbf{r}_{12})}{r_{12}^2} \right]$$

relativistic potential energy:

$$\begin{aligned} &\boldsymbol{\alpha}_1 \cdot \boldsymbol{\alpha}_2 \\ &\left[ \boldsymbol{\alpha}_1 \cdot \boldsymbol{\alpha}_2 - \frac{(\boldsymbol{\alpha}_1 \cdot \mathbf{r}_{12})(\boldsymbol{\alpha}_2 \cdot \mathbf{r}_{12})}{r_{12}^2} \right] \end{aligned}$$

-- Gaunt interaction

-- Breit interaction

# Terms in Hamiltonian which appear due to relativistic corrections

$$\mathbf{H}_e^{\text{mv}} = -\frac{1}{8m^3c^2} \sum_{i=1}^{N_{\text{elec}}} \mathbf{p}_i^4 \quad \rightarrow \quad \text{mass-velocity term, present also in the absence of the applied field}$$

## One-electron correction terms:

$$\begin{aligned} \mathbf{H}_e^{\text{Zeeman}} &= g_e \mu_B \sum_{i=1}^{N_{\text{elec}}} \left[ \mathbf{s}_i \cdot \mathbf{B}_i - \frac{1}{2mc^2} (\mathbf{s}_i \cdot \mathbf{B}_i) \mathbf{p}_i^2 \right] \\ \mathbf{H}_e^{\text{SO}} &= \frac{g_e \mu_B}{4mc^2} \sum_{i=1}^{N_{\text{elec}}} [\mathbf{s}_i \cdot (\mathbf{F}_i \times \mathbf{p}_i - \mathbf{p}_i \times \mathbf{F}_i)] \\ \mathbf{H}_e^{\text{Darwin}} &= \frac{1}{8m^2c^2} \sum_{i=1}^{N_{\text{elec}}} \nabla \cdot \mathbf{F}_i \end{aligned}$$

$B_i$  —applied magnetic field

$F_i$  —applied electric field

$s_i, p_i$  —spin and momenta of electron  $i$

## Two-electron correction terms:

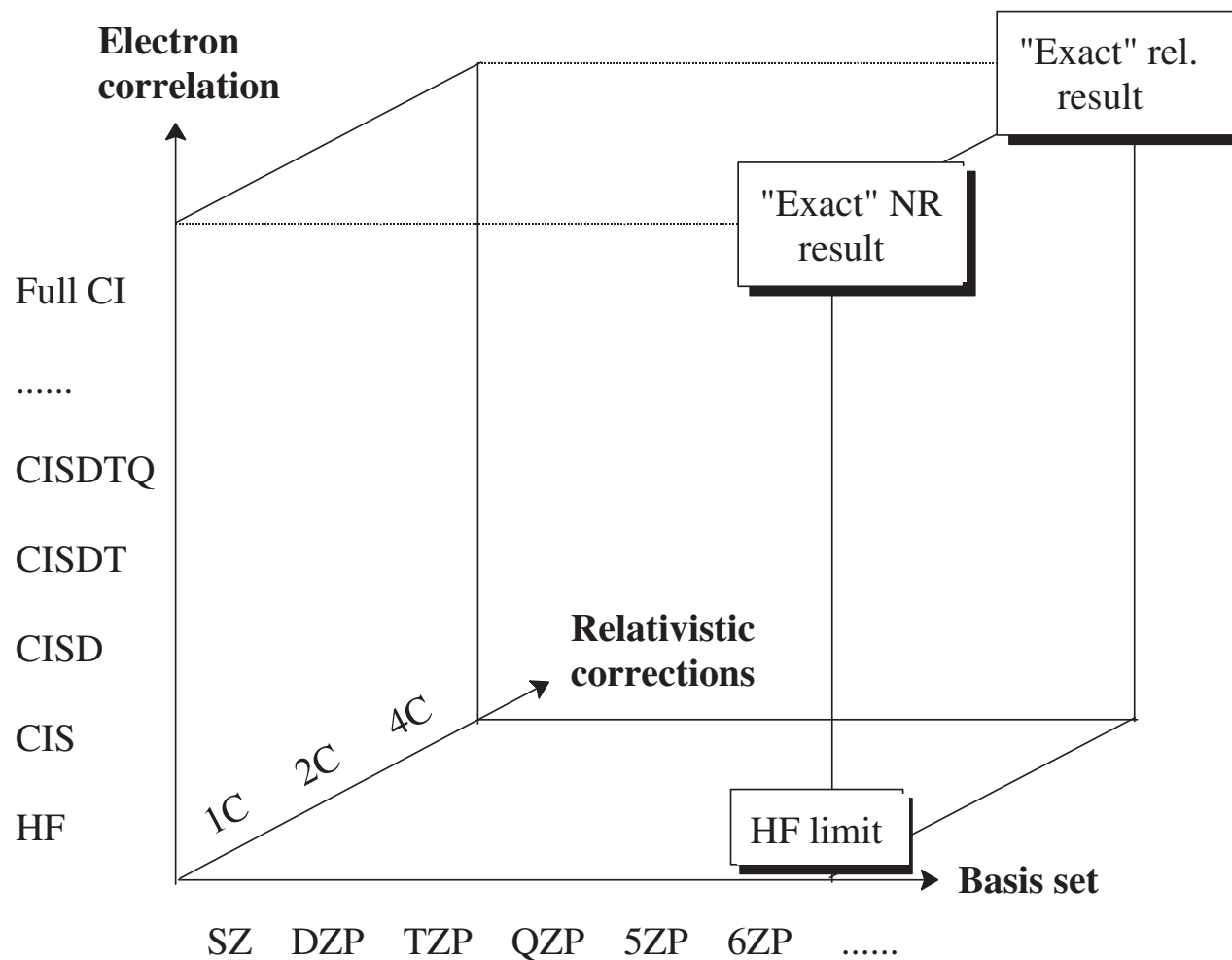
$$\begin{aligned} \mathbf{H}_{ee}^{\text{SO}} &= -\frac{g_e \mu_B}{2mc^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{j \neq i}^{N_{\text{elec}}} \frac{\mathbf{s}_i \cdot (\mathbf{r}_{ij} \times \mathbf{p}_i)}{r_{ij}^3} \\ \mathbf{H}_{ee}^{\text{SOO}} &= -\frac{g_e \mu_B}{mc^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{j \neq i}^{N_{\text{elec}}} \frac{\mathbf{s}_i \cdot (\mathbf{r}_{ij} \times \mathbf{p}_j)}{r_{ij}^3} \\ \mathbf{H}_{ee}^{\text{SS}} &= \frac{g_e^2 \mu_B^2}{2c^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{j \neq i}^{N_{\text{elec}}} \left( \frac{\mathbf{s}_i \cdot \mathbf{s}_j}{r_{ij}^3} - 3 \frac{(\mathbf{s}_i \cdot \mathbf{r}_{ij})(\mathbf{r}_{ij} \cdot \mathbf{s}_j)}{r_{ij}^5} - \frac{8\pi}{3} (\mathbf{s}_i \cdot \mathbf{s}_j) \delta(\mathbf{r}_{ij}) \right) \\ \mathbf{H}_{ee}^{\text{OO}} &= -\frac{1}{4m^2c^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{j \neq i}^{N_{\text{elec}}} \left( \frac{\mathbf{p}_i \cdot \mathbf{p}_j}{r_{ij}} + \frac{(\mathbf{p}_i \cdot \mathbf{r}_{ij})(\mathbf{r}_{ij} \cdot \mathbf{p}_j)}{r_{ij}^3} \right) \\ \mathbf{H}_{ee}^{\text{Darwin}} &= -\frac{\pi}{2m^2c^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{j \neq i}^{N_{\text{elec}}} \delta(\mathbf{r}_{ij}) \end{aligned}$$

# Terms in Hamiltonian which appear due to relativistic corrections

**Operators involving one electron and one nucleus:**

$$\begin{aligned}\mathbf{H}_{\text{ne}}^{\text{SO}} &= \frac{g_e \mu_B}{2mc^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{A=1}^{N_{\text{nuclei}}} Z_A \frac{\mathbf{s}_i \cdot (\mathbf{r}_{iA} \times \mathbf{p}_i)}{r_{iA}^3} \\ \mathbf{H}_{\text{ne}}^{\text{PSO}} &= \frac{\mu_N}{mc^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{A=1}^{N_{\text{nuclei}}} g_A \frac{\mathbf{I}_A \cdot (\mathbf{r}_{iA} \times \mathbf{p}_i)}{r_{iA}^3} \\ \mathbf{H}_{\text{ne}}^{\text{SS}} &= -\frac{g_e \mu_B \mu_N}{c^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{A=1}^{N_{\text{nuclei}}} g_A \left( \frac{\mathbf{s}_i \cdot \mathbf{I}_A}{r_{iA}^3} - 3 \frac{(\mathbf{s}_i \cdot \mathbf{r}_{iA})(\mathbf{r}_{iA} \cdot \mathbf{I}_A)}{r_{iA}^5} - \frac{8\pi}{3} (\mathbf{s}_i \cdot \mathbf{I}_A) \delta(\mathbf{r}_{iA}) \right) \\ \mathbf{H}_{\text{ne}}^{\text{Darwin}} &= \frac{\pi}{2m^2 c^2} \sum_{i=1}^{N_{\text{elec}}} \sum_{A=1}^{N_{\text{nuclei}}} Z_A \delta(\mathbf{r}_{iA})\end{aligned}$$

It is not practical to add all these terms in the Hamiltonian. Instead, these terms can be accounted by *perturbation*.



Converging the computational results by increasing the basis set, the amount of electron correlation and description of the relativistic effects.

# Solvent effects: overview

**Table 11. Solvation Energies (kcal/mol) for Some Solutes in Water (6-31G Calculations)**

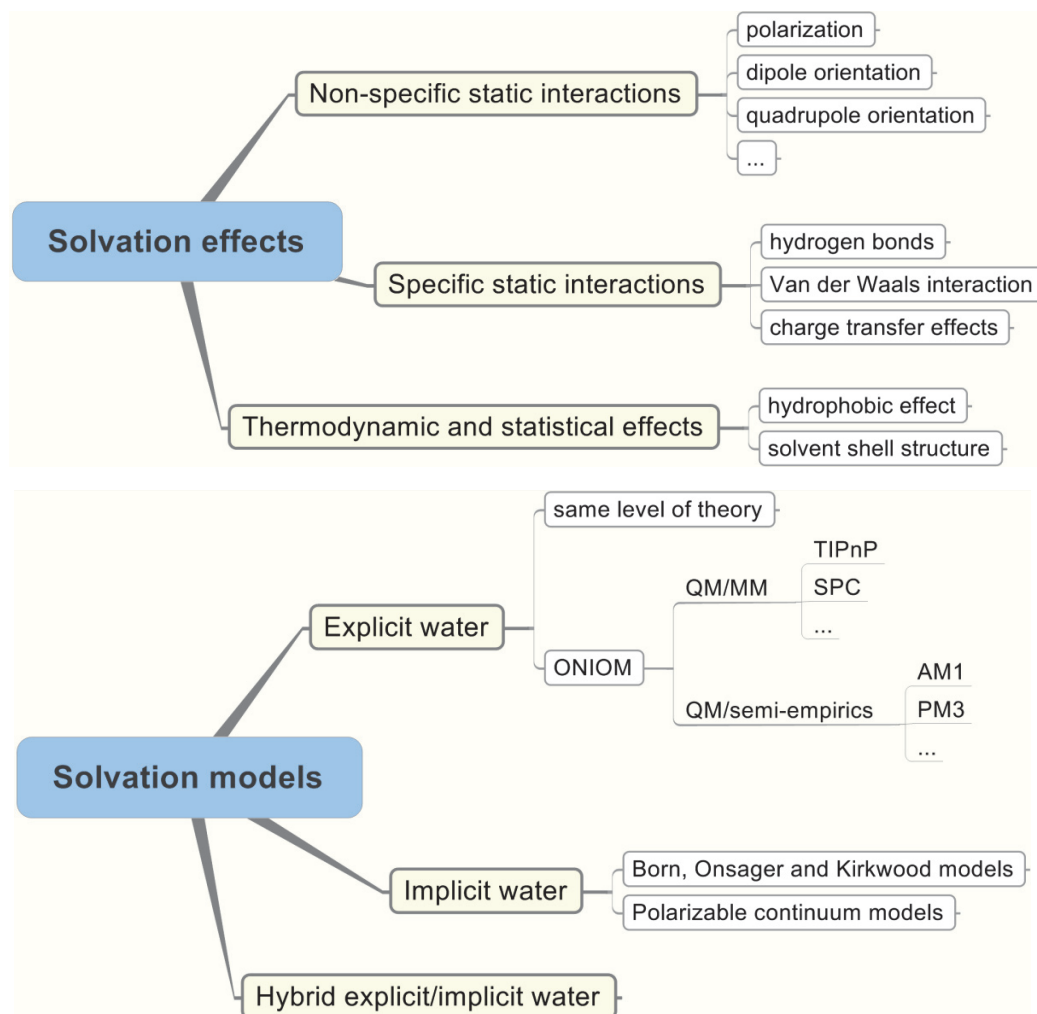
| solute           | method | $\Delta G_{\text{el}}^a$ | $\Delta G_{\text{dis}}^a$ | $\Delta G_{\text{el}} + \Delta G_{\text{dis}}^a$ | $\Delta G_{\text{el}} + \Delta G_{\text{dis}}^b$ |
|------------------|--------|--------------------------|---------------------------|--------------------------------------------------|--------------------------------------------------|
| H <sub>2</sub> O | SCF    | -9.14                    | -3.92                     | -13.06                                           | -13.09                                           |
|                  | PTE3   | -8.58                    | -4.23                     | -12.82                                           | -12.78                                           |
| NH <sub>3</sub>  | SCF    | -5.60                    | -4.88                     | -10.48                                           | -10.50                                           |
|                  | PTE3   | -5.41                    | -5.32                     | -10.73                                           | -10.63                                           |
| HF               | SCF    | -10.43                   | -3.18                     | -13.62                                           | -13.68                                           |
|                  | PTE3   | -9.72                    | -3.44                     | -13.16                                           | -13.22                                           |
| CH <sub>4</sub>  | SCF    | -0.57                    | -8.16                     | -8.73                                            | -8.65                                            |
|                  | PTE3   | -0.30                    | -9.01                     | -9.30                                            | -8.82                                            |

<sup>a</sup>  $\Delta G_{\text{el}}$  and  $\Delta G_{\text{dis}}$  separately computed from the same  $\phi_M$ , only  $\Delta G_{\text{el}}$  is included the effective Hamiltonian. <sup>b</sup> Both  $\hat{\mathcal{H}}_{\text{el}}$  and  $\hat{\mathcal{H}}_{\text{dis}}$  included in the effective Hamiltonian.

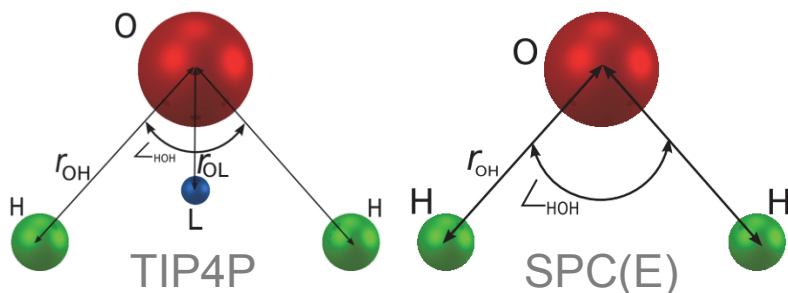
**Table 12. Barrier Height for the Methyl Torsion and Related Contribution to the Molecular Free Energy, Computed in the Hindered Rotor Approximation (kcal/mol) (4-31G SCF Calculations from Alagona et al.<sup>419</sup>)**

| molecule          |    | barrier height |                   | $-RT \ln(q_{\text{tors}})$ |                   |
|-------------------|----|----------------|-------------------|----------------------------|-------------------|
|                   |    | $\epsilon = 1$ | $\epsilon = 78.5$ | $\epsilon = 1$             | $\epsilon = 78.5$ |
| N-methylformamide | T  | 0.17           | 0.62              | -1.10                      | -0.83             |
|                   | C  | 0.67           | 0.80              | -0.87                      | -0.83             |
|                   | TS | 1.15           | 1.34              | -0.79                      | -0.76             |
| methyl formate    | T  | 0.71           | 1.09              | -0.93                      | -0.82             |
|                   | C  | 0.18           | 0.87              | -1.03                      | -0.87             |
|                   | TS | 1.14           | 1.10              | -0.79                      | -0.80             |

Many calculations, particularly with charged and polar molecules, essentially require a correct treatment of solvent electrostatics...



# Explicit water models: molecular dynamics



$$E_{\alpha\beta} = \epsilon_0 \left[ \left( \frac{\sigma_O}{r_{OO}} \right)^{12} - \left( \frac{\sigma_O}{r_{OO}} \right)^6 \right] + \sum_{nk} \frac{q_n q_k}{r_{nk}}$$

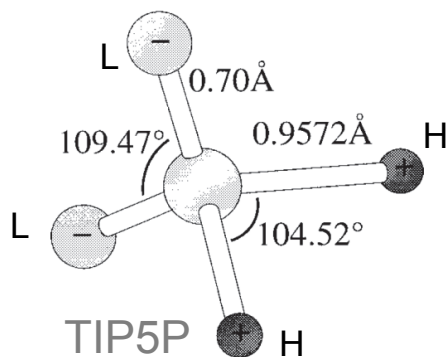


TABLE I. Monomer geometry, Lennard-Jones parameters, and partial charges describing the water-water interactions.

|                                   | SPC      | SPCE     | TIP4P    | TIP5P    |
|-----------------------------------|----------|----------|----------|----------|
| $q_H(e)$                          | +0.41    | +0.4238  | +0.520   | +0.241   |
| $q_O(e)$                          | -0.82    | -0.8476  | 0.0      | 0.0      |
| $q_L(e)$                          | ...      | ...      | -1.04    | -0.241   |
| $\epsilon_0(\text{kJ/mol})$       | 0.650 17 | 0.650 17 | 0.648 95 | 0.669 89 |
| $\sigma_O(\text{nm})$             | 0.316 56 | 0.316 56 | 0.315 36 | 0.312    |
| $r_{O-H}(\text{nm})$              | 0.0100   | 0.0100   | 0.095 72 | 0.095 72 |
| $r_{O-L}(\text{nm})$              | ...      | ...      | 0.015    | 0.070    |
| $\Theta_{\text{HOH}}(\text{deg})$ | 109.47   | 109.47   | 104.52   | 104.52   |
| $\Theta_{\text{LOL}}(\text{deg})$ | ...      | ...      | ...      | 109.47   |

TABLE III. Calculated and experimental properties for liquid water at 25 °C and 1 atm.

|                                            | TIP3P             | TIP4P             | TIP5P             | Expt. <sup>a</sup> |
|--------------------------------------------|-------------------|-------------------|-------------------|--------------------|
| $\rho \text{ (g/cm}^3\text{)}$             | $1.002 \pm 0.001$ | $1.001 \pm 0.001$ | $0.999 \pm 0.001$ | 0.997              |
| $-E \text{ (kcal/mol)}$                    | $9.82 \pm 0.01$   | $10.06 \pm 0.01$  | $9.87 \pm 0.01$   | 9.92               |
| $\Delta H_{\text{vap}} \text{ (kcal/mol)}$ | $10.41 \pm 0.01$  | $10.65 \pm 0.01$  | $10.46 \pm 0.01$  | 10.51              |
| $C_p \text{ (cal/mol deg)}$                | $20.0 \pm 0.6$    | $20.4 \pm 0.7$    | $29.1 \pm 0.8$    | 18.0               |
| $10^6 \kappa \text{ (atm}^{-1}\text{)}$    | $64 \pm 5$        | $60 \pm 5$        | $41 \pm 2$        | 45.8               |
| $10^5 \alpha \text{ (deg}^{-1}\text{)}$    | $92 \pm 8$        | $44 \pm 8$        | $63 \pm 6$        | 25.7               |

These models reproduce *structural* and *thermodynamic* properties of water.

SPC = Simple Point Charge  
TIP = Transferable Intermolecular Potential

# Explicit water models: molecular dynamics

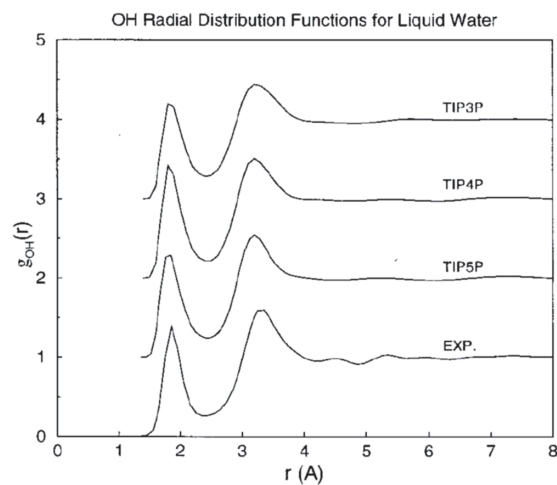
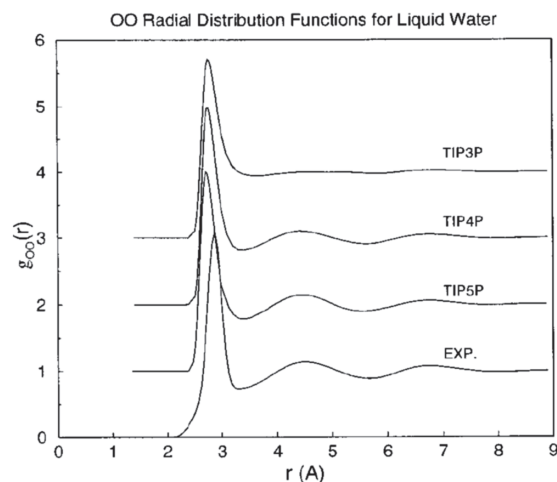


TABLE VI. Dipole<sup>a</sup> and quadrupole<sup>b</sup> moments for the water monomer.

|                    | $\mu$ | $Q_{xx}$ | $Q_{yy}$ | $Q_{zz}$ |
|--------------------|-------|----------|----------|----------|
| Expt. <sup>c</sup> | 1.85  | 2.63     | -2.50    | -0.13    |
| TIP3P              | 2.35  | 1.76     | -1.68    | -0.08    |
| TIP4P              | 2.18  | 2.20     | -2.09    | -0.11    |
| TIP5P              | 2.29  | 1.65     | -1.48    | -0.17    |

<sup>a</sup>Units are  $10^{-18}$  esu-cm.

<sup>b</sup>Units are  $10^{-26}$  esu-cm<sup>2</sup>.

TABLE V. Computed and experimental dielectric constants at 1 atm.

| $T$ (°C) | TIP5P          | Expt. <sup>a</sup> |
|----------|----------------|--------------------|
| 0.0      | $91.8 \pm 1.5$ | 87.74              |
| 25.0     | $81.5 \pm 1.6$ | 78.30              |
| 50.0     | $74.7 \pm 1.9$ | 69.91              |
| 75.0     | $68.6 \pm 1.8$ | 62.43              |
| 100.0    | $60.3 \pm 1.8$ | 55.72              |

Note excellent agreement with the experiment on the distance correlation function and considerable deviations from the experimental values for the electrostatic properties, particularly dipole moment.

# Explicit water models: semi-empirics and HF

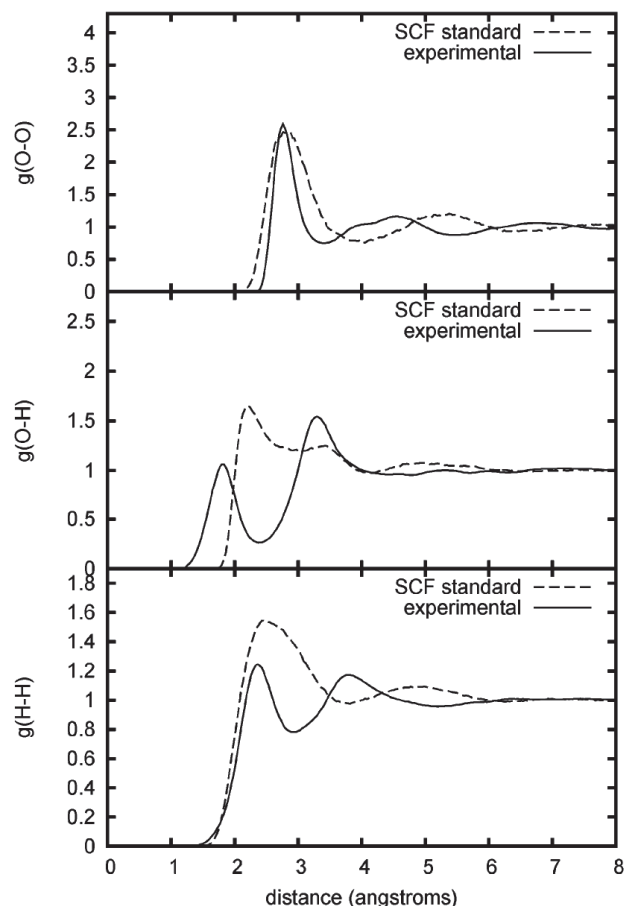


Figure 1. Radial distribution functions for liquid water obtained in the MD simulation with the AM1 method and experimental results.<sup>35</sup>

Semi-empirical methods fail to reproduce the structural properties of liquid water – even the first coordination shell is incorrect.

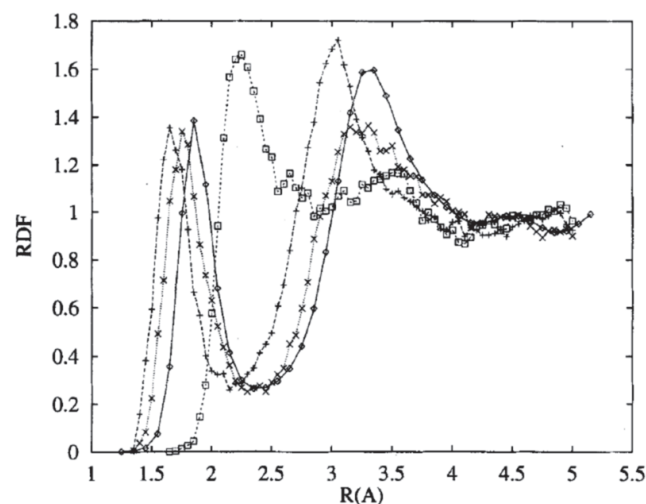


FIGURE 2 Radial distribution function  $g(r_{OH})$ . Experimental [11] ( $\circ$ ), classical MD ( $\times$ ), AM1 ( $\square$ ), HF ( $+$ ).

Hartree-Fock method is considerably better, on both the structural and the dielectric properties, but on the scale of solvated molecules it is very expensive.

# Continuum solvation effects

We will assume that the molecule sits in a cavity made in a polarisable material. The energy of the molecule has two contributions:

$$E = \underbrace{\langle \Psi | \hat{H} | \Psi \rangle}_{\text{energy of the molecule (incl. electrostatic interactions with the cavity)}} - \underbrace{\langle \Psi | \hat{G} | \Psi \rangle}_{\text{energy of cavity formation}}$$

**Table 11.1** Nucleic acid base dipole moments (D) at the SM5.42R/HF/6-31G(d) level

| Molecule     | Dipole moment |            |       |
|--------------|---------------|------------|-------|
|              | Gas           | Chloroform | Water |
| Adenine      | 2.4           | 2.9        | 3.1   |
| Cytosine     | 6.5           | 8.0        | 8.5   |
| Guanine      | 5.3           | 6.7        | 7.1   |
| Hypoxanthine | 6.4           | 7.8        | 8.2   |
| Thymine      | 4.4           | 5.6        | 6.0   |
| Uracil       | 4.5           | 5.6        | 6.0   |

## Cavity models:

1. **Kirkwood-Onsager**: assumes spherical cavity polarized by the dipole moment of the molecule.  
**Completely wrong for all practical purposes.**
2. **Polarisable continuum model (PCM)**: assumes the cavity to be a union of spheres centred on individual atoms. Quite good, but can exhibit numerical issues with tessellation, integration and geometry convergence.
3. **Isodensity PCM (IPCM)**: uses the SCF isodensity surface as the cavity. Repeats the SCF calculation until the cavity no longer changes.
4. **Self-consistent isodensity PCM (SCIPCM)**: updates the cavity within the SCF loop, achieving convergence for SCF and cavity at the same time.

F. Jensen, *Introduction to Computational Chemistry*, Wiley, 2007.

# Obsolete solvent cavity models

Born model refers to a spherical cavity with a charge placed inside:

$$\Delta G_{\text{electr}} = -\frac{\varepsilon - 1}{2\varepsilon} \frac{q^2}{R}$$

Bell model refers to a spherical cavity with a dipole placed inside:

$$\Delta G_{\text{electr}} = -\frac{\varepsilon - 1}{2\varepsilon + 1} \frac{\mu^2}{R^3}$$

Onsager model refers to a spherical cavity with a polarizable dipole:

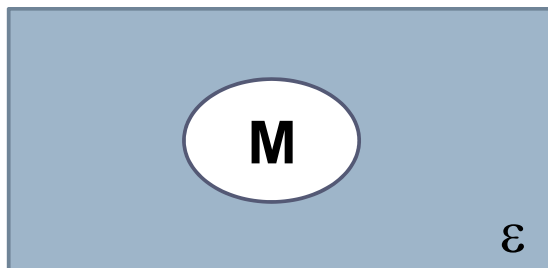
$$\Delta G_{\text{electr}} = -\frac{\varepsilon - 1}{2\varepsilon + 1} \frac{\mu^2}{R^3} \left[ 1 - \frac{\varepsilon - 1}{2\varepsilon + 1} \frac{2\alpha}{R^3} \right]$$

Kirkwood model refers to a general multipole expansion of the electrostatic potential of the molecule in a spherical and elliptical (Kirkwood-Westheimer) cavities.

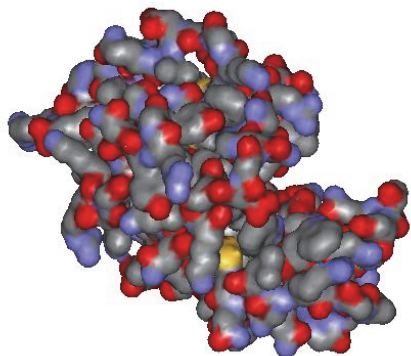
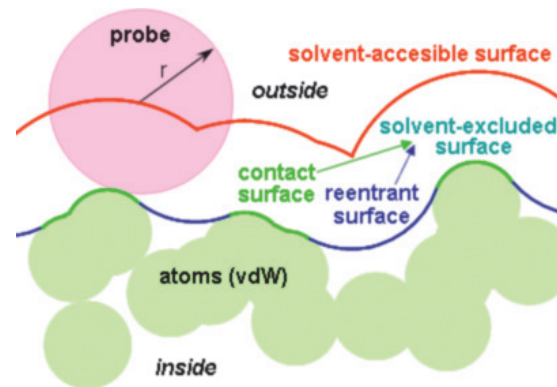
*Common feature of all four models: they are completely inaccurate for all practical purposes.*

F. Jensen, *Introduction to Computational Chemistry*, Wiley, 2007.

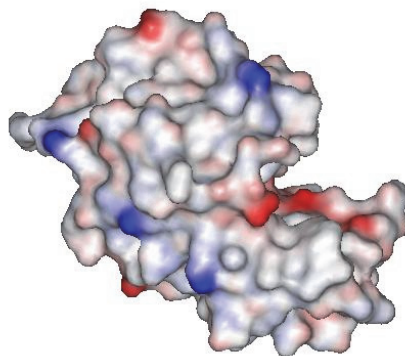
# Current solvent cavity models



$$\Delta G_{\text{solvation}} = \Delta G_{\text{cavity}} + \Delta G_{\text{dispersion}} + \Delta G_{\text{electrostatic}}$$



Van der Waals surface



Lee-Richards surface

Van der Waals surface is the surface of the union over VdW spheres of individual atoms. It is easier to generate and differentiate, and it is often preferred in quantum chemical calculations.

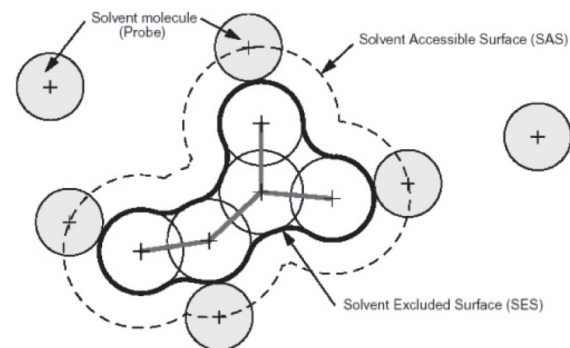
Solvent accessible surface (*Lee-Richards surface*) is obtained by rolling a probe sphere on a collection of atoms (represented by their Van der Waals radius) and tracing the surface containing its centre. Solvent excluded surface (*Connolly surface*) is obtained by tracing the closest point instead of the centre.

# Polarizable continuum model

The apparent surface charge is computed as the component of the potential gradient in the internal side of the surface, perpendicular to the surface.

$$\sigma(\vec{r}) = -\frac{\epsilon - 1}{4\pi\epsilon} \frac{\partial}{\partial \vec{n}} [\varphi(\vec{r})]_{\text{in}}$$

The resulting charges are fed into the quantum mechanical model (the details are very complicated, see the 2005 review by Tomasi, Mennucci and Cammi for details).



**Figure 1.** Solvent accessible surface (SAS) traced out by the center of the probe representing a solvent molecule. The solvent excluded surface (SES) is the topological boundary of the union of all possible probes that do not overlap with the molecule.

## Variations:

1. *Integral equation formalism (IEF) PCM:* (the default method in Gaussian03) uses Green's functions for the Poisson's equation to solve the electrostatic problem.
2. *Isodensity PCM (IPCM):* uses the SCF isodensity surface as the cavity. Repeats the SCF calculation until the cavity no longer changes.
3. *Self-consistent isodensity PCM (SCIPCM):* updates the cavity within the SCF loop, achieving convergence for SCF and cavity at the same time.

J. Tomasi, B. Mennucci, R. Cammi, <http://dx.doi.org/10.1021/cr9904009>

# Polarizable continuum model

The PCM model results depend on the choice of atomic radii used to generate the surface (Pauling radii are obtained from crystallographic data, Bondi radii also include other factors, Orozco-Luque radii have an exception for acidic hydrogens):

Free-energy of hydration ( $\Delta \mathcal{E}_{\text{solv}}$ ) in water calculated by PCM/DIR on G94 with different choices of radii and scaling factors (values in kcal/mol). In parentheses the errors with respect to experimental values

| Set of radii<br>Scaling factor               | Bondi<br>1.2  | Pauling<br>1.2 | Orozco-Luque<br>1.2 | Pauling<br>1.2, 1.0 | Orozco-Luque<br>1.25 | Experimental<br>$\Delta \mathcal{E}_{\text{solv}}$ |
|----------------------------------------------|---------------|----------------|---------------------|---------------------|----------------------|----------------------------------------------------|
| CH <sub>3</sub> COOH                         | -4.47 (+1.1)  | -5.96 (+0.7)   | -7.86 (-1.2)        | -7.59 (-0.9)        | -6.76 (-0.1)         | -6.7 <sup>a</sup>                                  |
| CH <sub>3</sub> COONH <sub>2</sub>           | -6.87 (+2.8)  | -8.51 (+1.2)   | -11.40 (-1.7)       | -10.76 (-1.1)       | -9.28 (+0.4)         | -9.7 <sup>a</sup>                                  |
| CH <sub>3</sub> COCH <sub>3</sub>            | -3.10 (+0.7)  | -3.92 (-0.1)   | -3.92 (-0.1)        | -3.92 (-0.1)        | -3.10 (+0.7)         | -3.8 <sup>a</sup>                                  |
| CH <sub>3</sub> Cl                           | -2.34 (-1.7)  | -2.59 (-2.0)   | -2.59 (-2.0)        | -2.59 (-2.0)        | -2.31 (-1.7)         | -0.6 <sup>a</sup>                                  |
| CH <sub>3</sub> CN                           | -4.74 (-0.8)  | -5.75 (-1.8)   | -5.75 (-1.8)        | -5.75 (-1.8)        | -5.06 (-1.2)         | -3.9 <sup>a</sup>                                  |
| CH <sub>3</sub> COOCH <sub>3</sub>           | -2.60 (+0.7)  | -3.59 (-0.3)   | -3.59 (-0.3)        | -3.59 (-0.3)        | -2.81 (+0.5)         | -3.3 <sup>a</sup>                                  |
| CH <sub>3</sub> NH <sub>2</sub>              | -1.96 (+2.6)  | -1.91 (+2.7)   | -3.68 (+0.9)        | -3.05 (+1.5)        | -3.09 (+1.5)         | -4.6 <sup>a</sup>                                  |
| CH <sub>3</sub> OH                           | -3.05 (+2.0)  | -3.50 (+1.6)   | -5.15 (0.0)         | -4.86 (+0.2)        | -4.39 (+0.6)         | -5.1 <sup>a</sup>                                  |
| CH <sub>2</sub> OH-CH <sub>2</sub> OH        | -4.24 (+3.5)  | -5.00 (+2.7)   | -7.60 (+0.1)        | -7.13 (+0.6)        | -6.23 (+1.5)         | -7.7 <sup>a</sup>                                  |
| NH <sub>3</sub>                              | -3.53 (+0.8)  | -3.69 (+0.6)   | -6.29 (-2.0)        | -5.39 (-1.1)        | -5.50 (-1.2)         | -4.3 <sup>a</sup>                                  |
| H <sub>2</sub> O                             | -4.99 (+1.3)  | -5.73 (+0.6)   | -8.84 (-2.5)        | -8.24 (-1.9)        | -8.18 (-1.9)         | -6.3 <sup>a</sup>                                  |
| average unsigned<br>error                    | 1.64          | 1.30           | 1.15                | 1.05                | 1.03                 |                                                    |
| CH <sub>3</sub> COO <sup>-</sup>             | -69.34 (+6)   | -74.92 (+0.1)  | -74.92 (+0.1)       | -74.92 (+0.1)       | -72.14 (+3)          | -75.0 <sup>b</sup>                                 |
| CH <sub>3</sub> NH <sub>3</sub> <sup>+</sup> | -67.67 (+0.3) | -68.21 (-0.2)  | -74.95 (-7)         | -73.94 (-6)         | -72.64 (-5)          | -68.0 <sup>b</sup>                                 |
| CH <sub>3</sub> PH <sub>3</sub> <sup>+</sup> | -63.25 (+0.8) | -63.21 (+0.8)  | -67.02 (-3)         | -66.04 (-2)         | -64.61 (-0.6)        | -64.0 <sup>b</sup>                                 |

<sup>a</sup> Ref. [29]. <sup>b</sup> Ref. [30].

J. Tomasi, B. Mennucci, R. Cammi, <http://dx.doi.org/10.1021/cr9904009>

# Polarizable continuum model

Although the SCIPCM method is undoubtedly the best, it tends to be *expensive*. The basis set effect is completely unpredictable – the solvation energy does not converge to the experimental value.

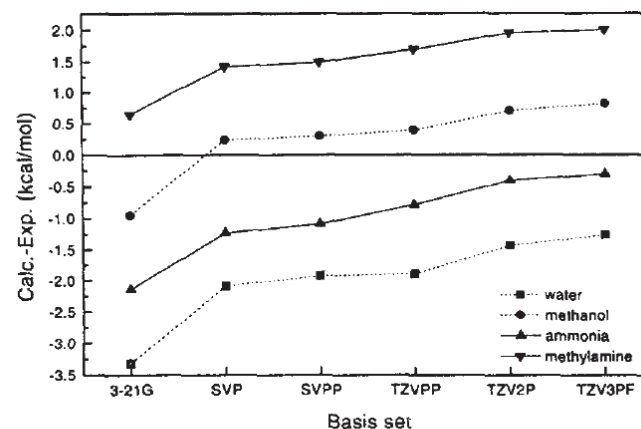
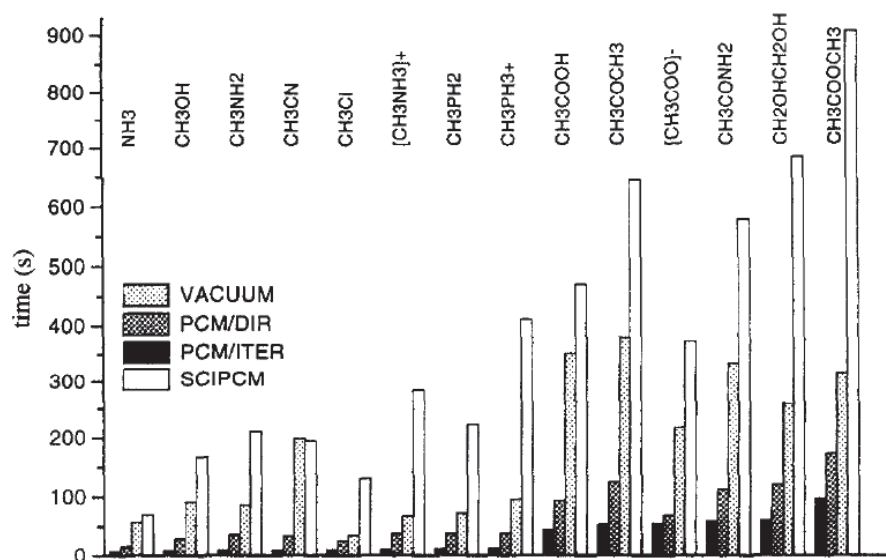
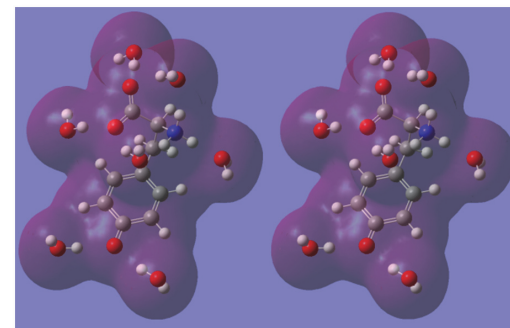
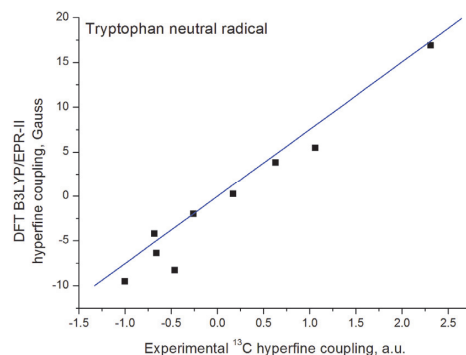
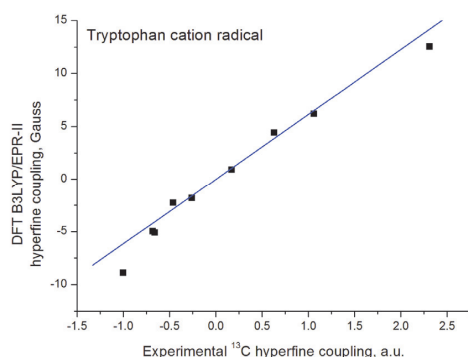


Fig. 2. Difference between calculated and experimental free-energy of solvation for  $\text{H}_2\text{O}$ ,  $\text{NH}_3$ ,  $\text{CH}_3\text{OH}$ ,  $\text{CH}_3\text{NH}_2$ , with different basis sets: SVP = 6-31G(d); SVPP = 6-31G(d, p); TZVPP = 6-311 + G(d, p); TZV2P = 6-311 + G(2d, 2p); TZV3PF = 6-311 + G(3df, 2p).

Convergence difficulties sometimes exhibited by Gaussian16 implementation of SCIPCM can be resolved by supplying a converged vacuum SCF orbitals as the initial guess for the solvated calculation.

# Explicit + PCM solvation

Accurate treatment of solvation is essential for the calculation of hyperfine couplings and  $g$ -factors in polar molecules.



It is often necessary to introduce the first level of solvation (such as hydrogen bonds) manually, by adding explicit solvent molecules – the effects of hydrogen bonding are not reproduced by continuum solvation models.

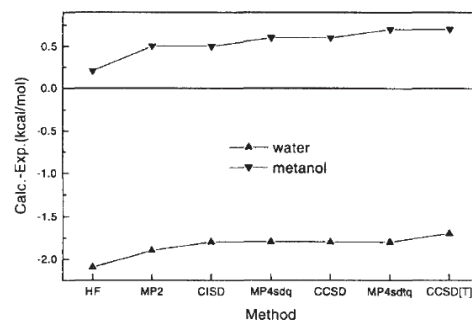


Fig. 3. Difference between calculated and experimental free-energy of solvation for  $\text{H}_2\text{O}$  and  $\text{CH}_3\text{OH}$  at different levels of calculation.

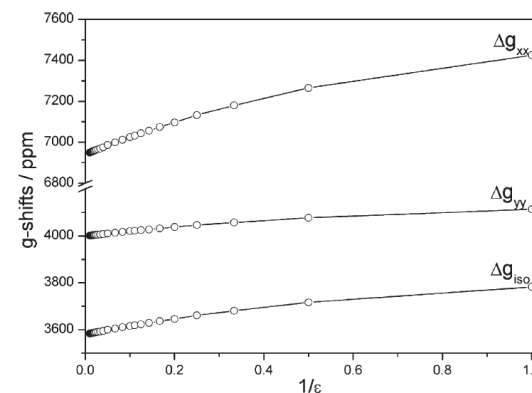
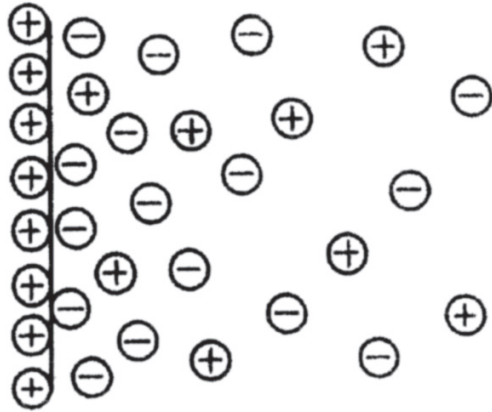


Figure 2. Calculated  $g$ -shifts of the DTBNO radical as a function of  $1/\epsilon$  with  $\epsilon$  being the dielectric constant (COSMO, B3LYP//B3LYP). The very small  $\Delta g_{zz}$  shifts ( $-191$  ppm for  $\epsilon = 1$  and  $-200$  ppm for  $\epsilon = 100$ ) are not displayed.

S. Sinnecker *et al.*, <http://dx.doi.org/10.1021/jp056016z>

# Poisson-Boltzmann model



The energy of an electrical charge is proportional to the local electrostatic potential:

$$E(\vec{r}) = ze\phi(\vec{r})$$

Therefore, from Boltzmann distribution:

$$n_+ = n_0 \exp\left(-\frac{ze\phi(\vec{r})}{kT}\right) \quad n_- = n_0 \exp\left(+\frac{ze\phi(\vec{r})}{kT}\right)$$

The local charge density is the difference between the concentrations of positive and negative charges:

$$\rho(\vec{r}) = ze(n_+ - n_-) = zen_0 \left\{ \exp\left[-\frac{ze\phi(\vec{r})}{kT}\right] - \exp\left[+\frac{ze\phi(\vec{r})}{kT}\right] \right\}$$

But the local charge density is, in its turn, related to the local electrostatic potential:

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi(\vec{r}) = -\frac{\rho(\vec{r})}{\epsilon_0 \epsilon}$$

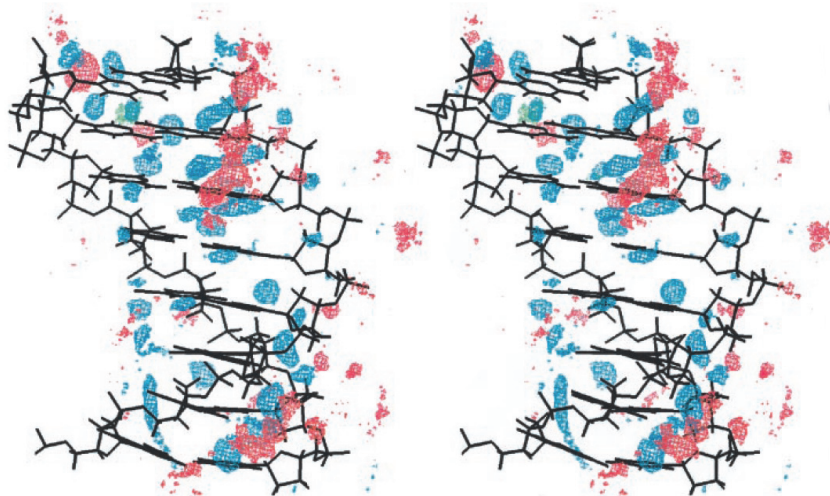
D.J. Shaw, *Colloid and Surface Chemistry*, Butterworth Heinemann, 2003.

# Poisson-Boltzmann model

If we combine the two equations, we get:

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi(\vec{r}) = -\frac{ze n_0}{\epsilon_0 \epsilon} \left\{ \exp \left[ -\frac{ze \phi(\vec{r})}{kT} \right] - \exp \left[ +\frac{ze \phi(\vec{r})}{kT} \right] \right\}$$

This is known as the Poisson-Boltzmann equation. Because it assumes the presence of independent point charges in the liquid, it is *not applicable to dielectric liquids*.



Cross-eyed stereo plot of water (blue), sodium (red) and chloride (green) ion distribution around a DNA molecule.

$$\Delta G_{\text{electr}} = \frac{1}{2} \int \rho(\vec{r}) \phi_{\text{reac}}(\vec{r}) d^3 \vec{r}$$

$$\phi_{\text{reac}}(\vec{r}) = \phi_{\text{solv}}(\vec{r}) - \phi_{\text{vac}}(\vec{r})$$

The equation is solved numerically on a 3D grid. The resulting reaction potential (the difference between the solvated and the vacuum potential) determines the electrostatic part of the Gibbs free energy of solvation.

D.J. Shaw, *Colloid and Surface Chemistry*, Butterworth Heinemann, 2003.

# Hydrogen bonds

Hydrogen bonds are primarily electrostatic and are described adequately by good DFT exchange-correlation functionals as well as post-HF methods:

**Table 12-1.** Deviation in the computed  $R_{O-O}$  distance of the water dimer [ $\text{\AA}$ ] from the 'experimental'  $R_e$  value of 2.952  $\text{\AA}$ .

| Basis Set     | HF    | MP2                 | CCSD(T)             | SVWN           | BLYP   | SLYP   | BVWN  | B3LYP  |
|---------------|-------|---------------------|---------------------|----------------|--------|--------|-------|--------|
| 6-31++G(d,p)  | 0.035 | -0.038              | -0.030              | -0.256         | -0.040 | -0.301 | 0.061 | -0.066 |
| 6-311++G(d,p) | 0.082 | -0.042              | -0.028              | -0.244         | -0.025 | -0.291 | 0.082 | -0.052 |
| aug-cc-pVDZ   | 0.080 | -0.034 <sup>a</sup> | -0.033 <sup>b</sup> | - <sup>c</sup> | -0.012 | -0.290 | 0.117 | -0.041 |
| aug-cc-pVTZ   | 0.086 | -0.044 <sup>a</sup> | -0.057 <sup>b</sup> | -0.241         | -0.004 | -0.288 | 0.124 | -0.034 |
| aug-cc-pVQZ   | 0.086 | -0.057 <sup>b</sup> | -                   | -0.238         | -0.003 | -0.286 | 0.119 | -0.033 |

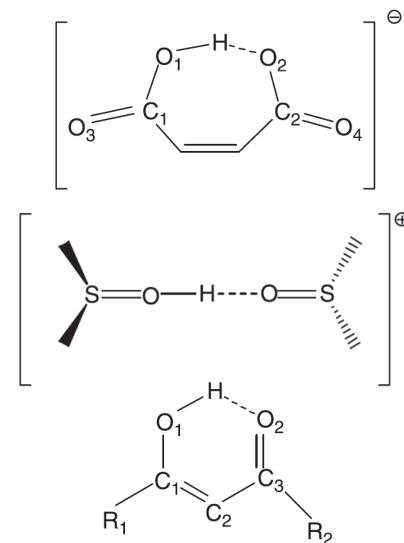
<sup>a</sup> Taken from Kim and Jordan, 1994. <sup>b</sup> Only  $R_{O-O}$  optimized, Halkier et al., 1997. <sup>c</sup> No  $C_s$ -*trans* minimum structure.

**Table 12-4.** Computed interaction energy ( $\Delta E$ ) and counterpoise-corrected interaction energies ( $\Delta E_{CP}$ ) of the water dimer [kcal/mol]. The BSSE is given in parentheses.

| Basis Set        | SVWN           |                 |        | BLYP       |                 |       | SLYP       |                 |        | BVWN       |                 |       |
|------------------|----------------|-----------------|--------|------------|-----------------|-------|------------|-----------------|--------|------------|-----------------|-------|
|                  | $\Delta E$     | $\Delta E_{CP}$ |        | $\Delta E$ | $\Delta E_{CP}$ |       | $\Delta E$ | $\Delta E_{CP}$ |        | $\Delta E$ | $\Delta E_{CP}$ |       |
| 6-31++G(d,p)     | 10.8           | 9.8             | (1.0)  | 5.6        | 4.8             | (0.8) | 13.2       | 12.3            | (0.9)  | 4.1        | 3.4             | (0.7) |
| 6-311++G(d,p)    | 10.5           | 9.5             | (1.0)  | 5.4        | 4.6             | (0.8) | 12.9       | 11.9            | (1.0)  | 4.0        | 3.3             | (0.7) |
| 6-311++G(3df,2p) | 9.3            | 9.1             | (0.2)  | 4.5        | 4.2             | (0.3) | 11.7       | 11.5            | (0.2)  | 3.1        | 2.9             | (0.2) |
| aug-cc-pVDZ      | - <sup>a</sup> |                 |        | 4.3        | 4.1             | (0.2) | 11.6       | 11.6            | (0.0)  | 2.9        | 2.8             | (0.1) |
| aug-cc-pVTZ      | 9.0            | 9.1             | (-0.1) | 4.2        | 4.2             | (0.0) | 11.4       | 11.6            | (-0.2) | 2.9        | 2.8             | (0.1) |
| aug-cc-pVQZ      | 9.0            | 9.1             | (-0.1) | 4.2        | 4.2             | (0.0) | 11.3       | 11.6            | (-0.2) | 2.9        | 2.9             | (0.0) |

<sup>a</sup> No  $C_s$ -*trans* minimum structure.

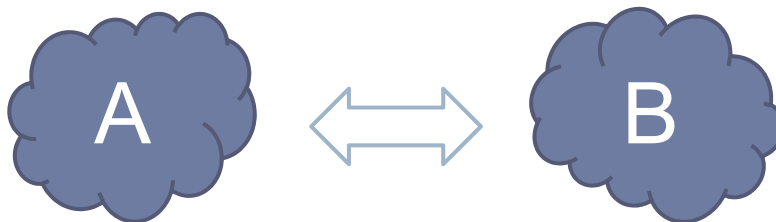
Strong (partially covalent) hydrogen bonds, such as the ones listed below, require high-level post-HF methods.



W. Koch, M.C. Holthausen, *A Chemist's Guide to Density Functional Theory*, Wiley, 2001.

# Non-covalent interactions

Van der Waals interaction and aromatic stacking are due to the two molecules polarizing each other on approach:



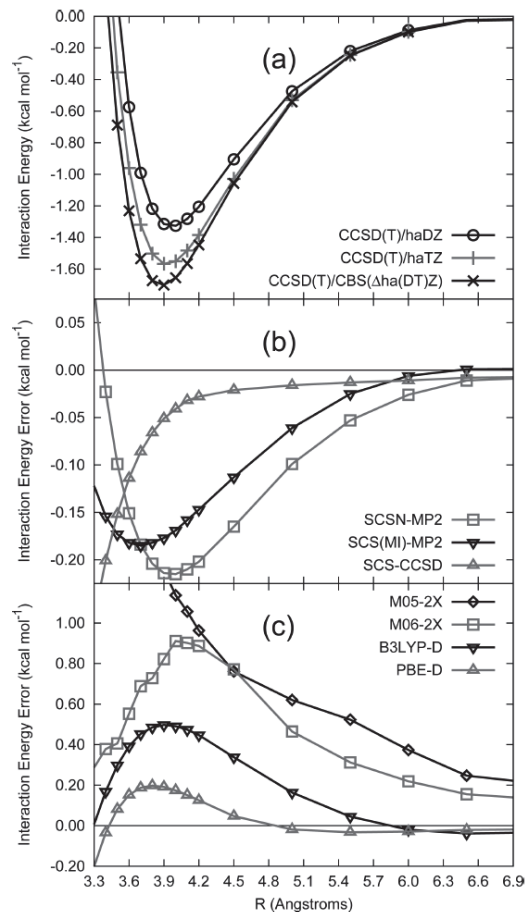
$$\hat{H} = \hat{H}_A + \hat{H}_B + \hat{H}_{AB}; \quad \hat{H}_{AB} = \sum_{a \in A} \sum_{b \in B} \frac{z_a z_b}{r_{ab}}$$

$$E = E_A + E_B + \underbrace{\langle 0_A 0_B | \hat{H}_{AB} | 0_A 0_B \rangle}_{\substack{\text{simple electrostatics -} \\ \text{not good enough}}} + \sum_{nk} \underbrace{\frac{\langle 0_A 0_B | \hat{H}_{AB} | n_A k_B \rangle \langle n_A k_B | \hat{H}_{AB} | 0_A 0_B \rangle}{E_0^{(A)} + E_0^{(B)} - E_k^{(A)} - E_n^{(B)}}}_{\substack{\text{requires the knowledge of} \\ \text{all excited determinants}}}$$

Unless the excited determinants are introduced accurately in a particular method, it is unlikely to yield accurate Van der Waals interactions.

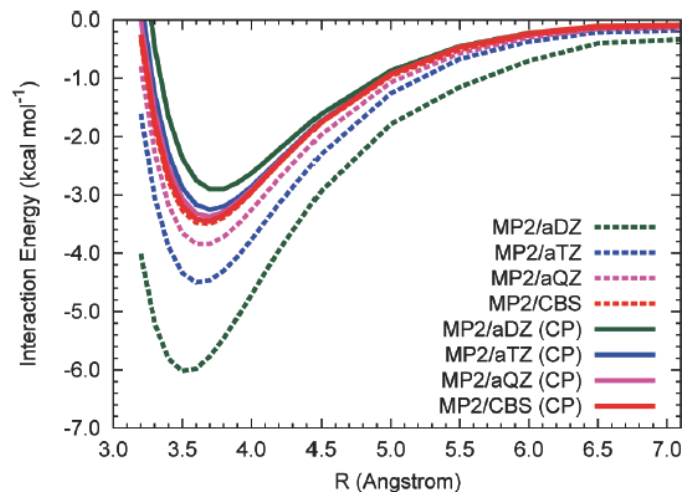
F. Jensen, *Introduction to Computational Chemistry*, Wiley, 2007.

# Non-covalent interactions



**Figure 6.** (a) CCSD(T) PECs for the sandwich benzene dimer. Interaction energy errors for (b) SCS-type approximations and (c) DFT methods compared to the CCSD(T)/CBS(Δha(DT)Z) interaction energies.

HF, DFT and other single-determinant methods completely fail to describe Van der Waals interactions and aromatic stacking – these interactions are almost entirely electron correlation effects.



**Figure 3.** Effect of counterpoise (CP) correction on MP2 potential energy curves for the sandwich benzene dimer.

MP2/TZ generally gives good results, but BSSE correction is essential for quantitative interpretation.

# UV/Vis spectra

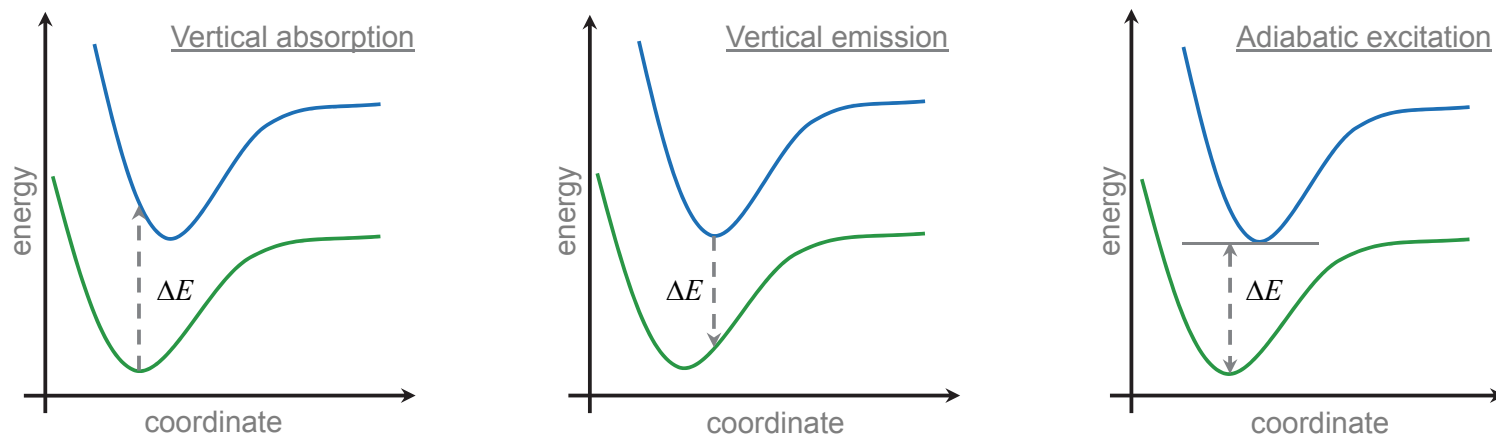
Accurate vertical excitation energies are usually obtained from configuration interaction calculations (excited determinants are expensive):

$$\Delta E_k = \langle \Psi_k | \hat{H} | \Psi_k \rangle - \langle \Psi_0 | \hat{H} | \Psi_0 \rangle$$

and the transition probabilities are calculated using perturbation theory:

$$\vec{d}_{kn} = \langle \Psi_k | \hat{\vec{d}} | \Psi_n \rangle \quad \hat{\vec{d}} = e\vec{r}$$

The energy minima for the ground and the excited states do not necessarily match:



F. Jensen, *Introduction to Computational Chemistry*, Wiley, 2007.

# UV/Vis spectra

## Notes on accuracy:

1. The energy accuracy is determined by the method used – CISD, CCSD and higher in a large basis are strongly preferred.
2. The line widths are determined by electronic structure *dynamics* and cannot, at present, be computed.
3. The transition moments are only qualitatively correct.
4. All the standard warnings about the lack of vibrational averaging, solvent effects, *etc.* apply.

TABLE II. Excitation energies  $\Delta E$  (eV), major contributing configurations, and % contributions of the leading double-excited configurations for the lowest  $\pi$  states of all-*trans* butadiene. Transition dipoles  $\mu$  (a.u.) from the ground state to  $nB_u$  states and from  $1B_u$  to  $nA_g$  states are also reported. All basis sets are DZp. The benchmark values of excitation energies obtained at CAS PT2/ANO are taken from Ref. [84].

| State  | CIS             |                | CIS(D)       |                 | SAC-CI (SD-R)   |                               |       | ATDA B3LYP   |                 | QR B3LYP              |              | CAS PT2 |
|--------|-----------------|----------------|--------------|-----------------|-----------------|-------------------------------|-------|--------------|-----------------|-----------------------|--------------|---------|
|        | $\Delta E$ (eV) | Configurations | $\mu$ (a.u.) | $\Delta E$ (eV) | $\Delta E$ (eV) | Configurations                | % $D$ | $\mu$ (a.u.) | $\Delta E$ (eV) | Configurations        | $\mu$ (a.u.) | /ANO    |
| $1B_u$ | 6.66            | (1-1')         | 2.661        | 6.77            | 6.86            | (1-1')                        | 0     | 2.291        | 5.94            | (1-1')                | 2.131        | 6.06    |
| $2B_u$ | 12.15           | (2-2')         | 0.784        | 11.33           | 11.72           | (2-2')+(11-1'2')              | 2     | 0.874        | 10.33           | (2-2')                | 0.706        |         |
| $3B_u$ |                 |                | -            | -               | 15.31           | (11-1'2')-(12-1'1')           | 78    | 0.17         | 14.51           | (1-1')((1-2')-(2-1')) |              |         |
| $2A_g$ | 9.66            | (1-2')+(2-1')  | 2.682        | 7.95            | 7.51            | (2-1')+(1-2')+(11-1'1')       | 13    | 1.695        | 7.11            | (2-1')+(1-2')         | 1.485        | 6.27    |
| $3A_g$ | 9.03            | (1-2')-(2-1')  | 0.112        | 9.28            | 9.41            | (1-2')-(2-1')                 | 0     | 0.306        | 8.57            | (1-2')-(2-1')         | 0.086        | 0.199   |
| $4A_g$ |                 |                | -            | -               | 12.65           | (2-1')+(1-2')-(11-1'1')       | 53    | 2.698        | 11.89           | (1-1')(1-1')          | 2.131        |         |
| $3A_g$ |                 |                | -            | -               | 16.71           | (12-1'2')+(22-1'1')+(11-2'2') | 83    | 0.557        | 16.28           | (1-1')(2-2')          | 0.706        |         |

Note the significant contributions from double excitations in many cases.

# Practical session

**Main Task :** Calculate the lowest excited state energy of the phenol ( $\text{C}_6\text{H}_5\text{-OH}$ ) dissolved in water.

## **Subtask 1: gas phase–ground geometry**

- 1) Open GaussView and build the molecular structure of phenol ( $\text{C}_6\text{H}_5\text{-OH}$ ).
- 2) Optimize the molecular structure using DFT (BP86), compute the vibrational frequencies and check if all of them are positive ( no imaginary frequencies present).

! BP86 def2-SVP TightSCF TightOpt Freq

- 3) Apply Time-Dependent DFT to optimize the first excited state.

! BP86 def2-SVP TightSCF KeepDens

%tddft

Nroots 2 # The number of excited states to calculate.

MaxDim 5

FollowIRoot 1

end

# Practical session

**Main Task :** Calculate the lowest excited state energy of the phenol ( $\text{C}_6\text{H}_5\text{-OH}$ ) dissolved in water.

## **Subtask 2: gas phase first-excited state geometry optimisation**

- 1) Setup the calculation to optimize the geometry to the first excited state.
- 2) Optimize the molecular structure using DFT (BP86), compute the vibrational frequencies and check if all of them are real and positive.

! BP86 def2-SVP TightSCF TightOpt Freq Keepdens

%tddft

Nroots 2 # The number of excited states to calculate.

MaxDim 5

IRoot 2 # Selecting the excited state (here no. 2) to perform a geometry optimization on.

FollowIRoot 1

end

- 3) Compare the obtained molecular geometries in **Subtask 1 and Subtask 2**.
- 4) Compare the energies of the first excited states obtained in **Subtask 1 and Subtask 2**.

# Practical session

**Main Task :** Calculate the lowest excited state energy of the phenol ( $\text{C}_6\text{H}_5\text{-OH}$ ) dissolved in water.

**Subtask 3: geometry optimization in the presence of the water as solvent (Conductor-like Polarizable Continuum Model, CPCM)**

**>> ground state optimization:**

- 1) Setup the calculation to optimize the geometry to the ground state using water as implicit solvent
- 2) Optimize the molecular structure using DFT (BP86), compute the vibrational frequencies and check if all of them are real and positive.

**>> excited state optimization:**

- 1) Setup the calculation to optimize the geometry to the first excited state using water as implicit solvent
- 2) Optimize the molecular structure using DFT (BP86), compute the vibrational frequencies and check if all of them are real and positive.

Compare the obtained geometries with the gas-phase geometries obtained in **Subtask 1 and Subtask 2**.

Compare the obtained energies of the first excited states with the gas-phase first excitations obtained in **Subtask 1 and Subtask 2**.

**!CPCM(solvent)    # ORCA input**

**You may use either Gaussian or ORCA for these tasks.**

**In Gaussian, the solvent is implied by SCRF keyword:**

<https://gaussian.com/scrf/>